Date: June 4, 1999

To: D. Neumann, P. Gallagher, A. Meyer, R. Dimeo

From: N. Maliszewskyj

RE: HFBS Software changes

Per Pat's request, this document is intended to serve as a description and record of proposed changes to the data acquisition software for the NG2 backscattering spectrometer. The most pressing needs in this regard include changing the primary data acquisition mode to histogram neutron events *vs*. time rather than velocity, the incorporation of a white beam monitor, and the recording of additional trending information by the core program.

I will give a brief description of the existing infrastructure in order to provide context for the changes proposed.

Existing Hardware

1. STR7137 Latch – reads position of phase space transform chopper, also generates periodic pulses (the "heartbeat") which provide the time base for the data acquisition system.

2. STR7201 Scaler – accumulates counts from neutron scattering events

3. OR ADC #1 – reads Doppler velocity signal from laser vibrometer

4. OR ADC #2 – reads Doppler displacement signal from laser vibrometer

5. STR8080 DSP – digital signal processor. The heart of the data acquisition system, responsible for building the data histogram.

6. HK4F – Motorola-based CPU running OS-9 operating system, responsible for "slow control" of VME data acquisition. Communication with HK4F (heurikon) is attained via Ethernet.

7. Koyo PLC – programmable logic controller which monitors all ancillary sensors and manages interlocks

Existing Software

1. *Motorola 96000 DSP code* – instructions for digital signal processor which dictate how the neutron events capture by the STR7201 scaler are stored in the data histogram

2. *Hfbsdaq* (HK4F) – "slow control" program running on OS-9. Hfbsdaq is responsible for initializing VME modules, downloading lookup tables and instructions to the DSP, starting and stopping the DSP, and transferring the data histogram to the Linux host computer.

3. *Hfbscore* (Linux) – autonomous program mediating all interactions of instrument users with hardware. The core program is responsible for monitoring the status of ancillary equipment (through its interaction with the PLC), and for managing data collection.

4. *Hfbs* (Linux) – user interface to core program. Provides facility for live display of data being collected.

5. *Bsview* (Linux) – simple program for display of data already saved to disk.

Changes in data acquisition modes

Over the past year, we have collected data in two primary modes, usually referred to as "standard" and "teach." Teach mode bins neutron events against current PST position. In our recent discussions, standard mode has also been referred to as "velocity binning," because the neutron events are directly binned against the measured velocity of the Doppler drive.

Rather than completely abandon the existing standard mode altogether, I propose that the current infrastructure be modified to permit the coexistence of three data acquisition modes:

- Teach mode – bin neutron events against PST position at the time of measurement

  Data histogram size: 20 detector channels $\times$ 4096 PST positions

Detector Channel	Assignment
0	Beam monitor counts
1-16	Detector counts
19	P(PST) sample histogram

- Time binning mode – bin neutron events against time with respect to Doppler drive pickup

  Data histogram size: 20 detector channels $\times$ 4096 time channels

  PST position lookup table determines whether neutron events are binned in histogram

  DAQ heartbeat of 30 μs provides time base for measurement. For slower Doppler speeds it will be necessary to perform a table lookup to insure that a complete "frame" of acquisition will fit within 4096 channels.

  Doppler shaft pickup signal resets the clock to T=0. The TTL signal must be converted to ECL and fed into one of the unused channels (17-19) in the STR7201 scaler.

  Because this mode assumes that the Doppler drive frequency is perfectly stable, it will be necessary to constantly monitor the Doppler pickup throughout the course of a measurement.

  The Doppler velocity signal will be read and recorded in the data histogram for diagnostic purposes.

  The mapping from time to energy (and accompanying manipulation of data) for both live and offline plotting will be the responsibility of the user interface and data display programs, respectively.

Detector Channel	Assignment
0	Beam monitor counts
1-16	Detector counts
17	Doppler velocity readings v(t)
19	P(t) sample histogram

- Velocity binning mode – bin neutron events against measured Doppler velocity

  Data histogram size: 20 detector channels × 4096 velocity channels

  PST position lookup table determines whether neutron events are binned in histogram

  Velocity channel values can be either raw 12-bit velocity value read from the ADCs or a "composite" value formed from the 11-bit velocity value and the sign bit from the displacement signal. For the present, we will use the former.

  Perform flight time offset correction to insure proper matching of Doppler velocity with neutron events.

Detector Channel	Assignment
0	Beam monitor counts
1	P(v) sample histogram for monitor
2-17	Detector counts
18	P(v) sample histogram for detectors

In each data acquisition mode, a complete measurement is segmented in a series of "cycles," allowing the core program to periodically save data, omitting data from cycles in which problems were detected (e.g., the Doppler drive frequency changed, or sample temperature varied unacceptably).

Changes in slow control

The considerable time overhead in starting/stopping data acquisition (in any mode) can be attributed in large part to the fact that all required lookup tables were downloaded to the DSP from the heurikon every time acquisition was started. To reduce this time, I propose that lookup tables be downloaded to the DSP only when either the slow control program is started, or when requested by the Linux core program.

Incorporation of additional beam monitors

In the coming cycle, HFBS will have three separate beam monitors: the original beam monitor mounted near the analyzer, a new fission chamber mounted downstream from the PST, and a white beam monitor. Events from the new fission chamber will be histogrammed with those from the detectors, while those from the other monitors will be accumulated in simple counters. We can use a Joerger VSC16 TTL scaler in the VME crate, managed by *hfbsdaq*, to accumulate these events and transfer them to the Linux host.

Changes in sensor monitoring

The PLC is already monitoring water and oil temperatures and pressures, the chamber vacuum, and the speeds of the Doppler drive and PST. The core program obtains the current readings from the PLC and periodically records them to a log file.

The core program will be modified to read the Doppler drive pickup frequency in units of revolutions per minute and to record this information to data files and hardware logs.

Changes in data file structure

Our ultimate goal is to write data files in the hierarchical data format (HDF), but in the mean time we have been writing data in a simple ASCII format. In either case, Andreas has requested that additional information be stored in data files. At a minimum, the following information will be stored:

- Comments – general purpose comments indicating sample and run conditions
- User – list of users who "own" the data file

- Run mode – data acquisition mode
- Start time – time run was started
- Stop time – time run completed
- Number of cycles – number of data acquisition cycles completed
- Cycle time – nominal time for completion of a single data acquisition cycle
- Livetime – total live acquisition time
- Doppler frequency – a record of Doppler drive pickup frequency readings (one for each completed cycle)
- Monitor sums – accumulated counts in white and secondary beam monitors
- Histogram – Neutron counts recorded from detectors (including P(v))

Schedule and priority of software changes

As the changes to the data acquisition infrastructure (how the raw data are managed, manipulated, and stored) are the most important, I intend to start work on the slow control and the portions of the core program responsible for controlling data acquisition and transferring data from the heurikon. Once that step has been accomplished, I will start working on the DSP code for the incorporation of the time binning and the (slightly modified) velocity binning modes.

Because the time binning mode depends on the absolute stability of the Doppler drive frequency, the next important task will be to incorporate the recording of this quantity into the core program.

Following this, I will incorporate monitoring of the white and secondary beam monitors into *hfbsdaq*. The logic signals from these monitors will be fed into our VSC16 TTL scaler.

A lower priority task is to coordinate the infrastructure changes with the Linux-side programs used to plot live and recorded data. As Rob has been independently developing an IDL-based program for display of data recorded on disk, updating *bsview* is of lower priority than updating *hfbs*. I consider it an important and useful feature of the data acquisition system that live data can be displayed in units of energy. In order to do this for data collected in time-binning mode, I will require an algorithm for performing the time/energy conversion and requisite data “folding” from Andreas and Rob.

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